

Solution Architecture

Project: FitBuddy – AI Fitness Plan Generator Using Gemini

Purpose: Project-track submission material adapted to the supplied AI/ML/Gen-AI template.

Layers

Presentation: HTML/Jinja2 templates, CSS and JavaScript. Application: FastAPI routes and request validation. AI integration: Gemini model client and prompt construction. Persistence: SQLAlchemy and configured database.

Request lifecycle

Browser form → FastAPI endpoint → validate request → call Gemini service → normalize/handle response → save record → render/return result.

Operational dependencies

Python runtime, installed packages, database access, network connectivity, and Gemini API credentials.

Architecture note

This document is a conceptual view based on the supplied project README and phase plan; refer to the actual source code for exact endpoint names and implementation details.